INCH-POUND

MIL-S-19500/617
8 March 1994

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, DUAL,
COMMON CATHODE OR ANODE CENTER TAP, ULTRAFAST, TYPES 1N6672 THRU 1N6674
AND 1N6672R THRU 1N6674R, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon, dual high voltage, ultrafast power rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-254AA isolated).

1.3 Maximum ratings. (per leg)

Types	V_{RWM} I_D = 0.2 mA dc	I_{FSM} 1/ t_p = 8.3 ms	I_F 1/2/3/ T_C = 100°C	t_{rr} 1/	$R_{\theta JC}$ 1/	$R_{\theta JA}$ 1/	T_{STG} and T_J
	V dc	A (pk)	A dc	ns	°C/W	°C/W	°C
1N6672, 1N6672R	300	150	15	35	2.0	40	-65 To +200
1N6673, 1N6673R	400						
1N6674, 1N6674R	500						

1/ Each individual diode.
2/ Derate linearly at 150 mA/°C from +100°C to +150°C.
3/ Total package current is limited to 30A dc

2. APPLICABLE DOCUMENTS

2.1 Government documents.

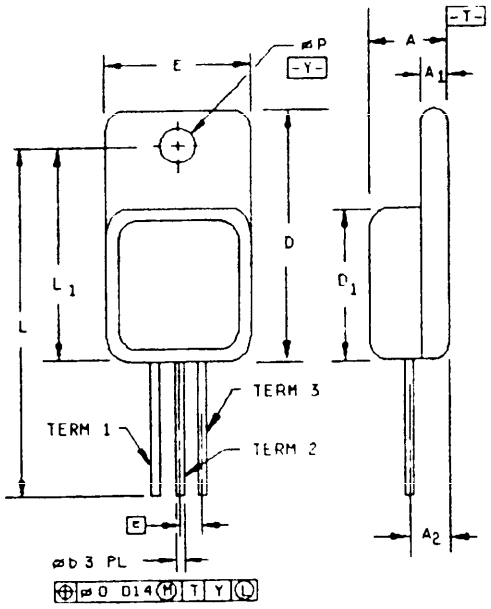
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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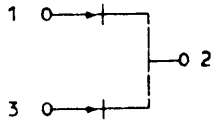
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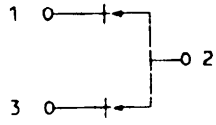
Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	6.32	6.60	.249	.260
A_1	1.02	1.27	.040	.050
A_2	3.81 BSC		.150 BSC	
D	20.07	20.32	.790	.800
D_1	13.59	13.89	.535	.545
e	3.81 BSC		.150 BSC	
E	13.59	13.89	.535	.545
L	30.35	31.37	1.195	1.235
L_1	16.89	17.40	.665	.685
øP	3.53	3.78	.139	.149
øb	0.89	1.43	.035	.045

SCHEMATIC

1N6672, 1N6673, 1N6674
TERM 1 = ANODE 1
TERM 2 = CATHODE
TERM 3 = ANODE 2



1N6672R, 1N6673R, 1N6674R
TERM 1 = ANODE 1
TERM 2 = CATHODE
TERM 3 = ANODE 2



NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. All terminals are isolated from case.

FIGURE 1. Dimensions and configuration (TO-254AA).

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The US government's preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead formation and finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14, of MIL-S-19500.

3.3.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Construction verification. Cross sectional photos from 3 devices shall be submitted in the qualification report.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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mounted remote to the generator package

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c 1/	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	Not applicable	Not applicable
10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	MIL-STD-750, method 1038; test condition A; t = 240 hours V_R = 80% of rated V_R; T_A = 125°C	MIL-STD-750, method 1038; test condition A; t = 48 hours V_R = 80% of rated V_R; T_A = 125°C
13	Subgroup 2 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±0.1 V (pk); ΔI_{R1} = ±10 µA dc or 100% from the initial value, whichever is greater.	Subgroup 2 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±0.1 V (pk); ΔI_{R1} = ±10 µA dc or 100% from the initial value, whichever is greater.

1/ Thermal impedance shall be performed any time after screen 3.

4.3.1 Burn-in conditions. Burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). V_R = 0.8 to 0.85 rated V_R dc (see 1.3)

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable $\bar{X}$, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record date $Z_{\theta JX}$). $Z_{\theta JX}$ shall be supplied on one lot (500 pieces minimum and a thermal response curve shall be submitted. Twenty two of these samples shall be serialized and provided to the qualifying activity for correlation prior to shipment of parts. Measurements conditions shall be in accordance with 4.4.1.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measure current - - - - - - - - - - - - - - - - 10 mA
b. I_H forward heating current - - - - - - - - - - - 15 - 50 A
c. t_H heating time - - - - - - - - - - - - - - - - - - 50 ms
d. t_{MS} measurement delay time - - - - - - - - - - - 100 µs minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}(max) = 1.6°C/W$.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions of tables IVa (JANS) and IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS of MIL-S-19500).

Subgroup	Method	Condition
2	1037	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). I_F or I_O = 1.25 to 10 A. Minimum for 2,000 cycles.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV of MIL-S-19500).

Subgroup	Method	Condition
3	1037	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). I_F or I_O = 1.25 to 10 A. Minimum for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
2	2036	Test condition A, weight = 10 pounds, t = 15 seconds.
6	1037	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). I_F or I_O = 1.25 to 10 A for 6,000 cycles.

4.4.4 Group E inspection. Group e inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.4.1 Group E inspection, table VII of MIL-S-19500.

Subgroup	Method	Condition	Sampling Plan
1	1051	500 cycles.	22 devices, c = 0
2	1038	Condition A, t = 1,000 hours V_R = 80% V_{rated}	22 devices, c = 0
3	Not applicable		
4	3101	T_A = +25°C; $R_{\theta JC}$ = rated $R_{\theta JC}$ (see 1.3).	5 devices, c = 0
5	1016	V_R = rated V_R (see 1.3); pressure = 33 mmHg; t = 1 minute (minimum), R_{ISO} = 2.0×10^6 ohm max.	5 devices, c = 0

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.6	°C/W
Breakdown Voltage	4022	IR = 500 µA dc pulsed (see 4.5.1)	VBR			Vdc
		1N6672, 1N6672R		300		
		1N6673, 1N6673R		400		
		1N6674, 1N6674R		500		
Forward voltage	4011	I_F = 10 A (pk); pulsed (see 4.5.1)	VF_1		1.35	V dc
		I_F = 20 A (pk); pulsed (see 4.5.1)	VF_2		1.55	V dc
Reverse current leakage	4016	DC method; V_R = 80% rated V_R (see 1.3); pulsed (see 4.5.1)	I_{R1}		50	µA dc
Subgroup 3						
High temperature operation		T_C = +100°C				
Reverse current leakage	4016	DC method; V_R = 80% rated V_R (see 1.3); pulsed (see 4.5.1)	I_{R2}		5	mA dc
Low temperature operation		T_A = -65°C				
Forward voltage	4011	I_F = 10 A (pk); pulsed (see 4.5.1)	V_{F3}		1.45	V dc
Subgroup 4						
Scope Display Eval.	2/					
Reverse recovery time	4031	Condition B; I_F = .5A, I_R = 1A, IRR = .25A	t_{rr}		35	ns
Junction Capacitance	4001	V_R = 10v dc, f = 1 MHz VSIG = 50mV (p-p) (max)	CJ		150	pf
Subgroups 5, 6, & 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

2/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100µA/division and 50 to 100V/division. Reverse current over the knee shall be at least 500µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in paragraph 4.3.2.1 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation and finish as specified (see 3.3.1).

c. Product assurance level and type designation.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

Agent:
DLA - ES

(Project 5961-1636-02)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/617	2. DOCUMENT DATE (YYMMDD) 94/03/08

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, DUAL, COMMON CATHODE OR ANODE CENTER TAP, ULTRAFAST, TYPES 1N6672 THRU 1N6674 AND 1N6672R THRU 1N6674R, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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